Exp: 9
22/8/20

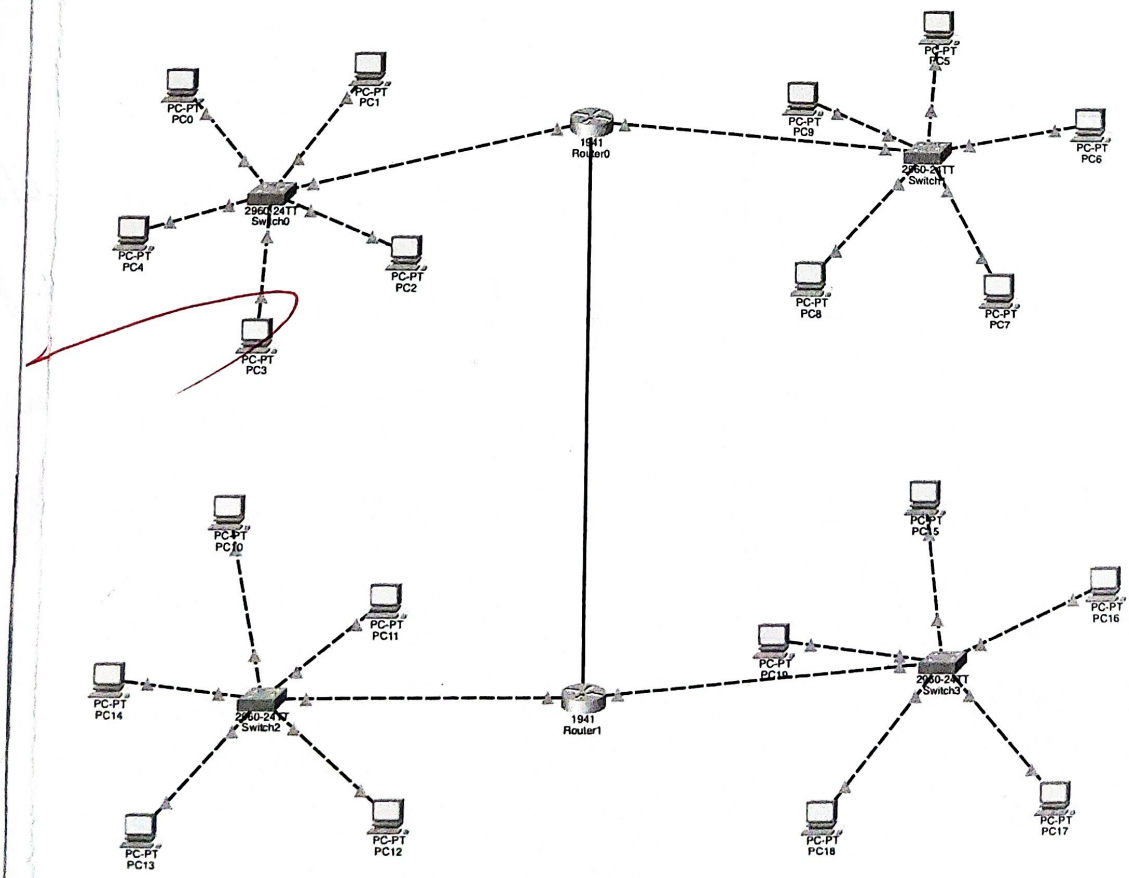
Implementation of Subnetting

Aim:

To implement subnetting in CISCO Packet Tracer.

Introduction:

Classless IP subnetting is a technique that allows for more effecient use of IP addresses by allowing for subnet masks that are not just the default masks for each IP class.



Steps for creating a Subnetting:

i) The first step is to create a network topology.

ii) To create a topology in Packet Tracer, select the 'New' button in the top left corner, then select 'Network' and 'Generic'.

iii) Once & after creating a topology, we can add devices.

iv) Add devices by dragging them from the bottom left corner into the topology.

v) Then connect them by dragging cable from one device's port to another device's port.

Subnetting:

To subnet the network address of 192.168.1.0/24 to provide enough space for atleast 5 addresses for end devices, the switch, and the router, we can use a /27 subnet mask. This will give us a 8 subnets with 30 host addresses each.

The IP addressing for the network shown in the topology can be as follows:

- Router R1:
- GigabitEthernet0/0: 192.168.1.1
- GigabitEthernet0/1: 192.168.2.1
- Switch S1:
- FastEthernet0/1: 192.168.1.0/27
- PC1: 192.168.1.11
- PC2: 192.168.1.12
- PC3: 192.168.1.13
- PC4: 192.168.1.14
- PC5: 192.168.1.15
- FastEthernet0/2: 192.168.2.0/27
- PC1: 192.168.2.11
- PC2: 192.168.2.12
- PC3: 192.168.2.13
- PC4: 192.168.2.14
- PC5: 192.168.2.15
- Router R2:
- FastEthernet0/0: 192.168.3.1
- FastEthernet0/1: 192.168.4.1
- Switch S2:
- FastEthernet0/1: 192.168.3.0/27
- PC1: 192.168.3.11
- PC2: 192.168.3.12
- PC3: 192.168.3.13
- PC4: 192.168.3.14
- PC5: 192.168.3.15
- FastEthernet0/2: 192.168.4.0/27
- PC1: 192.168.4.11
- PC2: 192.168.4.12
- PC3: 192.168.4.13
- PC4: 192.168.4.14
- PC5: 192.168.4.15

Configuring the Devices:

i) Configuring Router:

* Right-click on the router and select 'CLI'.
* In the CLI (Command Line Interface), enter the following commands.

enable
Configure terminal
interface Fast Ethernet 0/0
ip address {IP address} {subnet mask}
no shutdown
exit

* The first interface, Fast Ethernet 0/0, will be connected to the switch, ~~with~~ while the second interface, Fast Ethernet 0/1, will be connected to one of the PCs.

* Next configure the switch, by right clicking the switch and select CLI. Enter the following commands:

enable
configure terminal
interface Fast Ethernet 0/1
switchport mode access
exit
interface Fast Ethernet 0/2
switchport mode access
exit

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Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
	Successful	PC4(2)	Router2	ICMP		0.000	N	12
	Successful	PC4(2)	PC2(1)(1)	ICMP		0.000	N	13
	Successful	PC0	Router0	ICMP		0.000	N	14

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iii)

Result:

The implementation of Subnetting has been completed successfully.